BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MULTAN

INTERMEDIATE (12TH CLASS) EXAMINATION, 2024

BIOLOGY

Past Paper 2024 • Paper Code: 4461 / 4462

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

BIOLOGY PAPER-II GROUP-I (Objective Type)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 4461

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill that bubble in front of that question number, on bubble sheet. Use marker or pen to fill the bubbles. Cutting or filling two or more bubbles will result in zero mark in that question.

Q1. Which of the given is recovered in the collecting duct of the nephron?

(A) Glucose
(B) Water
(C) NaCl
(D) Potassium ions

Q2. The type of muscle having regular striations multinucleate and voluntary is:

(A) Skeletal muscle
(B) Smooth muscle
(C) Cardiac muscle
(D) All types of muscles

Q3. Cyclic activity of cross bridges is regulated by:

(A) Calcium ions
(B) Troponin
(C) ATP
(D) Actin

Q4. Given are the principle action of insulin except:

(A) Increasing glycogen synthesis
(B) Increasing cell utilization of glucose
(C) Inhibits hydrolysis of glycogen
(D) Promotes hydrolysis of glycogen

Q5. Cell bodies of sensory neurons constitute:

(A) Dorsal root ganglion
(B) Gray matter
(C) Ventral root ganglion
(D) Posterior root ganglion

Q6. Mature sperms are formed from spermatids through:

(A) Meiosis-I
(B) Meiosis-II
(C) Differentiation
(D) Mitosis

Q7. The head can be regenerated in:

(A) Earthworm
(B) Frog
(C) Leech
(D) Grasshopper

Q8. Which of the given is a stop codon?

(A) UUG
(B) UGA
(C) UCU
(D) CCA

Q9. To code 50 amino acids in a polypeptide chain, what will be the minimum number of nucleotides in its gene?

(A) 50
(B) 150
(C) 100
(D) 51

Q10. Which of the given is trisomy syndrome?

(A) Down's
(B) Edward
(C) Patau
(D) All of these

Q11. Different alleles of a gene that are both expressed in heterozygous condition are called:

(A) Complete dominance
(B) Incomplete dominance
(C) Codominant
(D) Over dominance

Q12. Which of the given is incorrectly matched?

(A) Protoplast – plant cell engineering
(B) RFLPs – DNA finger printing
(C) DNA polymerase – PCR
(D) DNA Ligase – Mapping humans chromosomes

Q13. Taq polymerase is used in PCR because of its:

(A) Low thermal stability
(B) High thermal stability
(C) High fidelity
(D) High speed

Q14. Lyell published the principles of:

(A) Geology
(B) Population
(C) Genetics
(D) Ecology

Q15. Diseases in living organisms which are caused by parasites are termed as:

(A) Mutualism
(B) Commensalism
(C) Infestations
(D) Succession

Q16. Coniferous forest located at high latitude are called:

(A) Alpine
(B) Boreal
(C) Taiga
(D) Prairies

Q17. The decline in thickness of ozone layer is caused by increasing level of:

(A) Hydrocarbon
(B) Nitro carbon
(C) Chlorine
(D) Chlorofluorocarbon

BIOLOGY PAPER-II GROUP-I (Subjective Type)

Time Allowed: 2.40 Hours | **Maximum Marks:** 68

> Note: Write same question number and its parts number on answer book, as given in the question paper.

Q. SECTION-I

Q2. (Marks 16) Attempt any eight parts. 8 × 2 =

(i) [1+1] How metanephridium is better than protonephridium?

(ii) [2] Categorise the plants distribution on the basis of osmoregulation.

(iii) [2] How can you describe blubber?

(iv) [1+1] Compare Epinasty with Hyponasty?

(v) [2] How would you define sliding filament model?

(vi) [2] How does jet propulsion mechanism work?

(vii) [1+1] What are advantages of Sexual Reproduction?

(viii) [2] How menstrual cycle is defined?

(ix) [2] Mention role of light in Limnetic zone.

(x) [1+1] Compare Coniferous alpine and Boreal forests.

(xi) [2] Define Greenhouse effect.

(xii) [2] Write any two sources of water pollution.

Q3. (Marks 16) Attempt any eight parts. 8 × 2 =

(i) What are the elements of nervous system?

(ii) Which factors control secretion of Antidiuretic hormone or Vasopressin?

(iii) Define Habituation. Give example.

(iv) Differentiate between Homozygote and Heterozygote.

(v) What are multiple Alleles? Give example.

(vi) How does sex determination occur in birds?

(vii) How can gene of interest be obtained?

(viii) What are the applications of PCR amplification and analysis?

(ix) Mention forensic application of DNA analysis.

(x) What is Biosphere?

(xi) Define Food web. Give its importance.

(xii) Write a note on Limnetic zone.

Q4. (Marks 12) Attempt any six parts. 6 × 2 =

(i) How do environmental factors contribute to abnormal development?

(ii) Why growth pattern in plants is called an open growth?

(iii) What are Fixed alleles?

(iv) How can you differentiate between Homologous and Analogous organs?

(v) Why do DNA replication always proceeds 5′ → 3′ directions?

(vi) What is a Point Mutation? Give one example.

(vii) How do different chromosomes differ from each other?

(viii) How are cancerous cells distinguished from normal cells?

(ix) Is interphase a resting phase? Why?

Q. SECTION-II

QNOTE. (Marks 3 × 8 = 24) Attempt any three questions.

Q5.

(a) [4] Explain different methods of excretion in plants.

(b) [4] What is Meiosis? Discuss prophase-I of meiosis in detail.

Q6.

(a) [4] Define Joints. How they are classified? Explain.

(b) [4] Define Succession? Explain Xerosere in detail.

Q7.

(a) [4] What is Synapse? How impulse can pass through synapse? Discuss it with suitable diagram.

(b) [4] Define Endangered species. Explain three measures to save endangered species.

Q8.

(a) [4] What are autosomes and sex-chromosomes? Explain sex-determination in humans.

(b) [4] Discuss the role of phytochromes in photoperiodism.

Q9.

(a) [4] Explain embryonic induction in detail.

(b) [4] What are transgenic bacteria? Write down their practical use in various fields.

GROUP – II

BIOLOGY PAPER-II GROUP-II (Objective Type)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 4462

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill that bubble in front of that question number, on bubble sheet. Use marker or pen to fill the bubbles. Cutting or filling two or more bubbles will result in zero mark in that question.

Q1. Which one of these requires least amount of water for its elimination from body?

(A) Creatinine
(B) Uric acid
(C) Ammonia
(D) Urea

Q2. There are _____ muscles present in human body.

(A) 650
(B) 630
(C) 680
(D) 206

Q3. The action of venus flytrap is called:

(A) Nytinasty
(B) Photonasty
(C) Thermonasty
(D) Haptonasty

Q4. Nociceptors in our body are related with:

(A) Vibration
(B) Touch
(C) Pain
(D) Light

Q5. Which one of these hormones promotes flowering in pineapple?

(A) Auxins
(B) Ethene
(C) Abscisic acid
(D) Cytokinins

Q6. Which of these is present between uterus and vagina?

(A) Urinogenital duct
(B) Oviduct
(C) Cervix
(D) Fallopian tube

Q7. During chick development, nervous system arises from:

(A) Eetoderm
(B) Mesoderm
(C) Endoderm
(D) Coelom

Q8. In sickle cell anemia, valine is present in hemoglobin in place of:

(A) Praline
(B) Glutamine
(C) Glutamic acid
(D) Isoleucine

Q9. X-Ray diffraction analysis of DNA was performed by:

(A) Erwin Chargaff
(B) Rosalind Franklin
(C) Watson and Crick
(D) Frederick Miescher

Q10. In yeast, cell cycle is completed in:

(A) 9 hours
(B) 10 hours
(C) 4.5 hours
(D) 1.5 hours

Q11. Which one of these syndromes is rare X-linked recessive trait?

(A) Color blindness
(B) Testicular feminization
(C) Hemophilia
(D) Hypophosphatemic rickets

Q12. Bacterial cells take up recombinant DNA when treated with:

(A) $CaCl_2$
(B) DNA ligase
(C) RNA polymerase
(D) Bacteriophage

Q13. Luciferase enzyme is found only in:

(A) Fruit fly
(B) Dragon fly
(C) Fire fly
(D) Butterfly

Q14. The main cause of extinction of species is:

(A) Pollution
(B) Habitat destruction
(C) Global warming
(D) Parasitism

Q15. The relationship between insects and flowering plants is an example of:

(A) Mutualism
(B) Parasitism
(C) Commensalisms
(D) Predation

Q16. Which zone is rich in life in aquatic ecosystem?

(A) Profundal zone
(B) Limnetic zone
(C) Littoral zone
(D) All of these

Q17. Stone cancer is a result of _____ pollution.

(A) Water
(B) Soil
(C) Sound
(D) Air

BIOLOGY PAPER-II GROUP-II (Subjective Type)

Time Allowed: 2.40 Hours | **Maximum Marks:** 68

> Note: Write same question number and its parts number on answer book, as given in the question paper.

Q. SECTION-I

Q2. (Marks 16) Attempt any eight parts. 8 × 2 =

(i) How does aldosterone play its role in concentration of urine?

(ii) What is special or unique feature of Malpighian tubules in insects?

(iii) Why does temperature of body increase during fever?

(iv) What is "All or None" response in muscle contraction?

(v) How does exercise affect a muscle?

(vi) How is pulvinus involved in sleep movements?

(vii) How are identical twins produced?

(viii) What do you know about the term oviparity?

(ix) Write down any two properties of hydrospheric ecosystem.

(x) Differentiate between Prairies and Savanna grasslands.

(xi) How are solid wastes useful in overcoming energy crisis?

(xii) Mention causes of Beriberi and Haemophilia.

Q3. (Marks 16) Attempt any eight parts. 8 × 2 =

(i) Define Habituation. Give two examples.

(ii) Write the role of a hormone in regulation of bile and pancreatic juice secretion.

(iii) Give the functions of sympathetic nervous system.

(iv) What are compound sex chromosomes? Write one example.

(v) Differentiate the sex-determination pattern in humans and birds.

(vi) What are Pseudoautosomal genes? Give one example.

(vii) What are Transgenic bacteria? Give their role in cleaning up beaches.

(viii) How Transgenic bacteria are better than Transgenic animals?

(ix) What is meristem culture? Write its one advantage.

(x) Draw a flow sheet of an energy pyramid showing transfer of energy from producers to tertiary consumers.

(xi) Differentiate between Primary and Secondary Succession.

(xii) Define the terms habitat and niche.

Q4. (Marks 12) Attempt any six parts. 6 × 2 =

(i) Highlight the role of morphogenetic determinants during development of an individual.

(ii) What do you know about discoidal cleavage?

(iii) "Genetic code is universal but not quite universal". Justify this statement.

(iv) How is lagging strand synthesized in the replication process?

(v) What is point mutation? Give one example.

(vi) Why interphase is called resting phase?

(vii) How is Phragmoplast formed? Give its importance for future daughter cells.

(viii) What are Hydrothermal vents?

(ix) Differentiate between Divergent and Convergent evolution.

Q. SECTION-II

QNOTE. (Marks 3 × 8 = 24) Attempt any three questions.

Q5.

(a) [4] Explain thermoregulatory strategies in mammals.

(b) [4] Describe Necrosis and Apoptosis.

Q6.

(a) [4] Explain process of repair of broken bones.

(b) [4] Write a note on Xerosere.

Q7.

(a) [4] What is a Nerve Impulse? Discuss the major factors involved in Resting Membrane Potential.

(b) [4] Discuss major points of Darwin's theory of natural selection.

Q8.

(a) [4] Enlist the names of different types of asexual reproduction in animals. Explain Parthenogenesis and its types.

(b) [4] Write a note on Erythroblastosis foetalis.

Q9.

(a) [4] Discuss the role of nucleus in development by giving the example of Acetabularia?

(b) [4] Write a detailed note on Gene sequencing.